

Ryan & Jaylen Rough Draft

Written Report

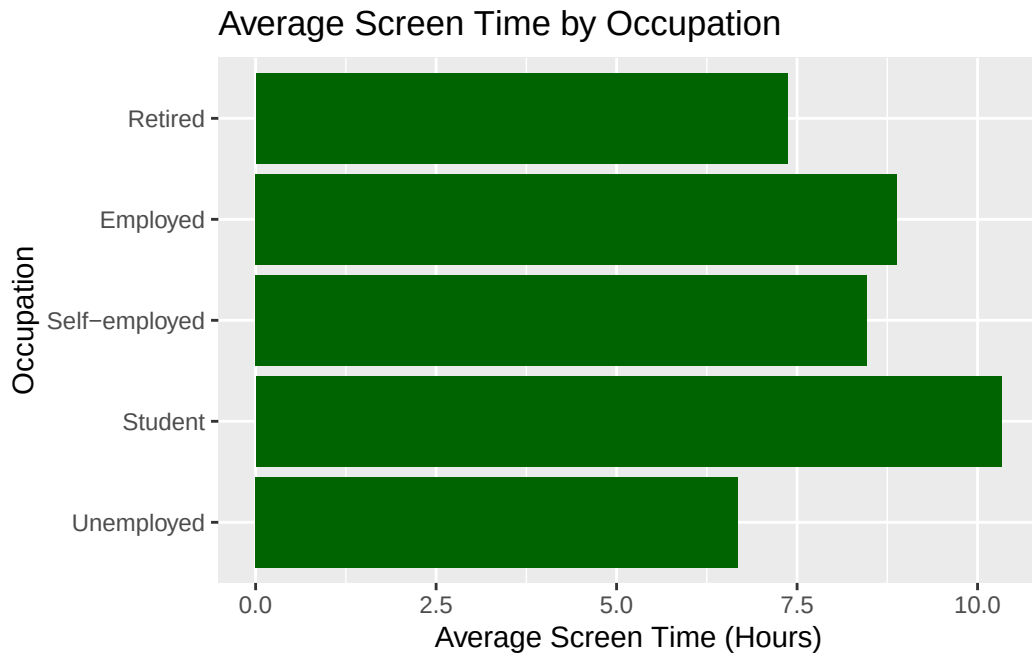
Introduction

Today, screens dominate our lives. People live with phones in their pockets, computers on their desks, and televisions in their homes. We sought to investigate the impact that screen time has on our lives, through metrics like reported productivity, mental wellness, stress, and sleep quality.

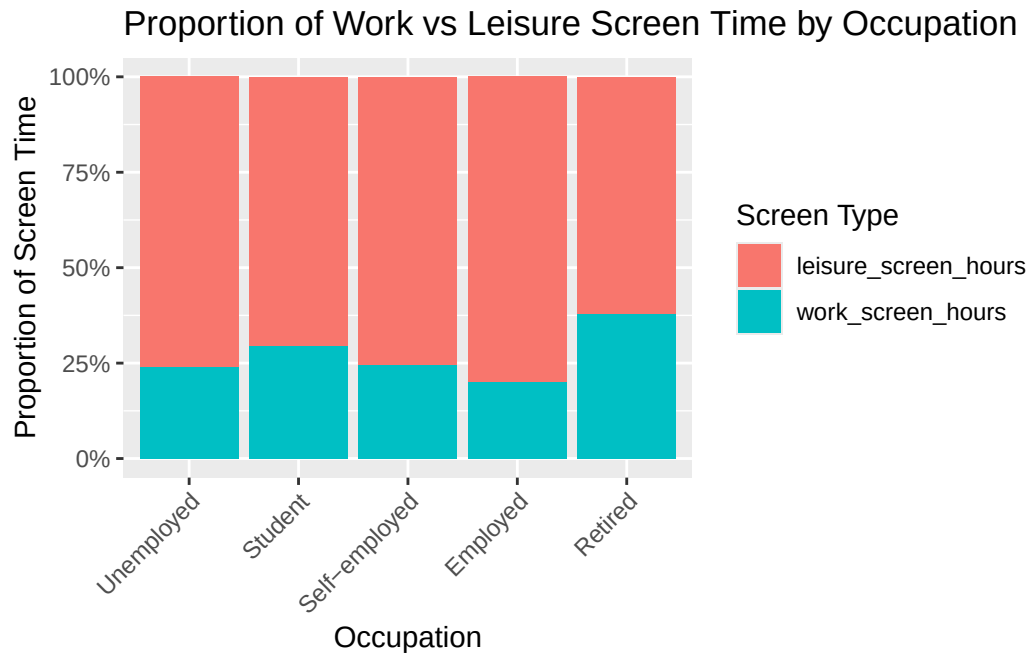
Exploratory Data Analysis

Who Uses Screens the Most?

We begin by examining which occupational groups accumulate the highest average screen time, as this helps us identify the populations most at risk. The graph below shows that students have the highest average screen time, followed by employed and then self-employed individuals.



To understand whether screen time skews more toward work or leisure within each occupation, we analyzed the proportional breakdown below. Counter intuitively, employed individuals have a higher percentage of leisure screen time, while retired individuals have a larger percentage of work screen time, comparatively. While we didn't have additional data to investigate this relationship, it provides interesting background and context.



How Do Work and Leisure Screen Time Relate?

Next, we examine whether individuals who spend more time on screens for work also tend to spend more time on screens for leisure, or whether the two types of usage trade off against each other.

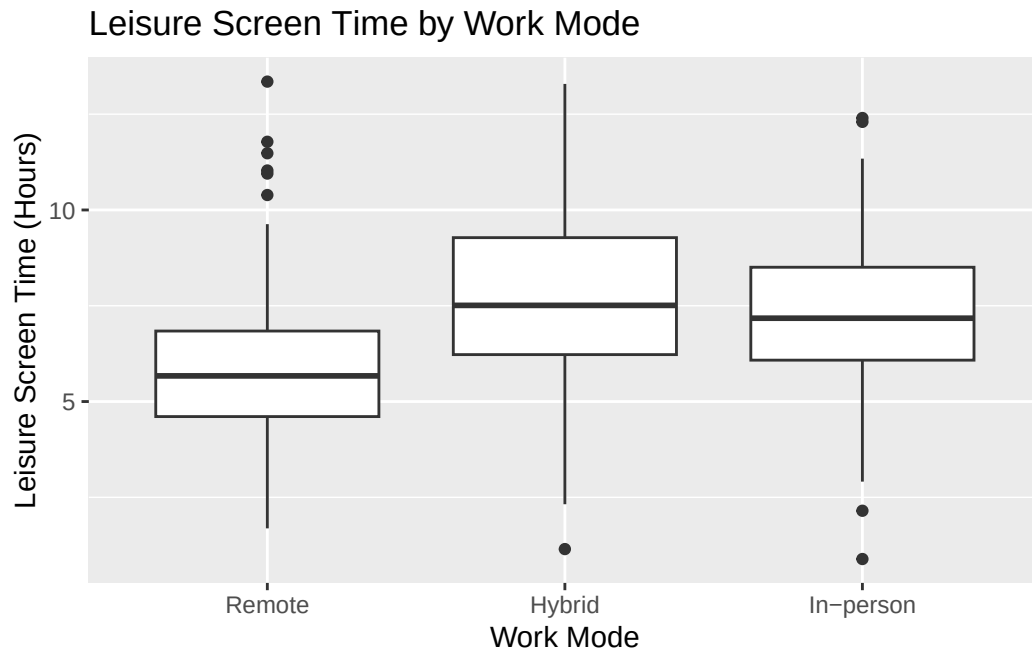
```
`geom_smooth()` using formula = 'y ~ x'
```

Work Screen Time vs Leisure Screen Time



Does Work Mode Affect Leisure Screen Time?

We also consider whether a person's work mode (remote, hybrid, or in-person) influences how much leisure screen time they accumulate outside of work.

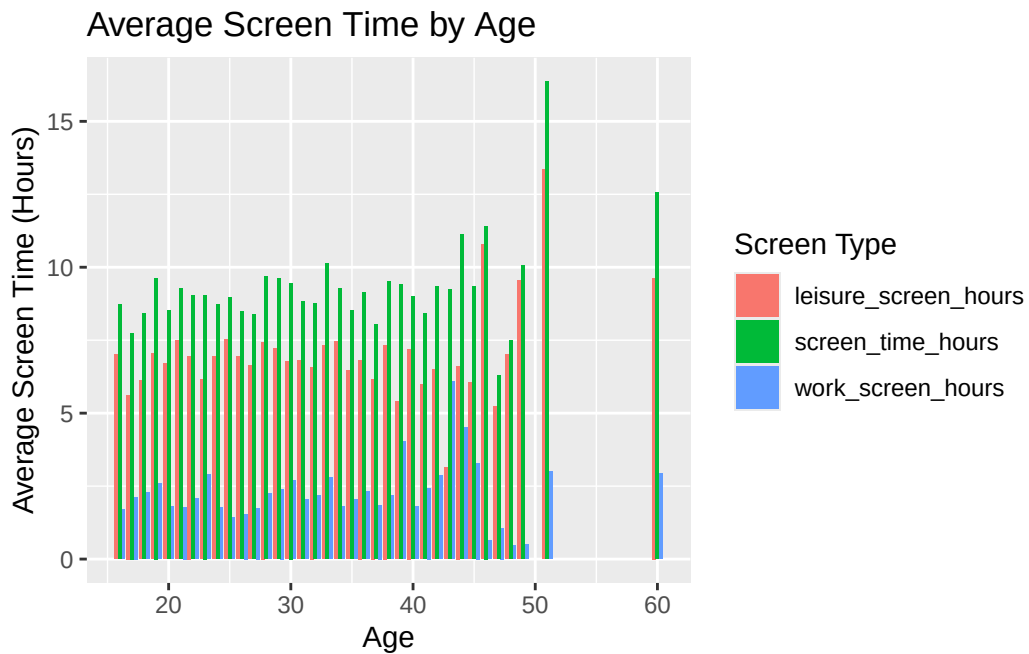


Does Screen Time Vary with Age?

We examine how total screen time is distributed across ages to identify any age-related patterns.



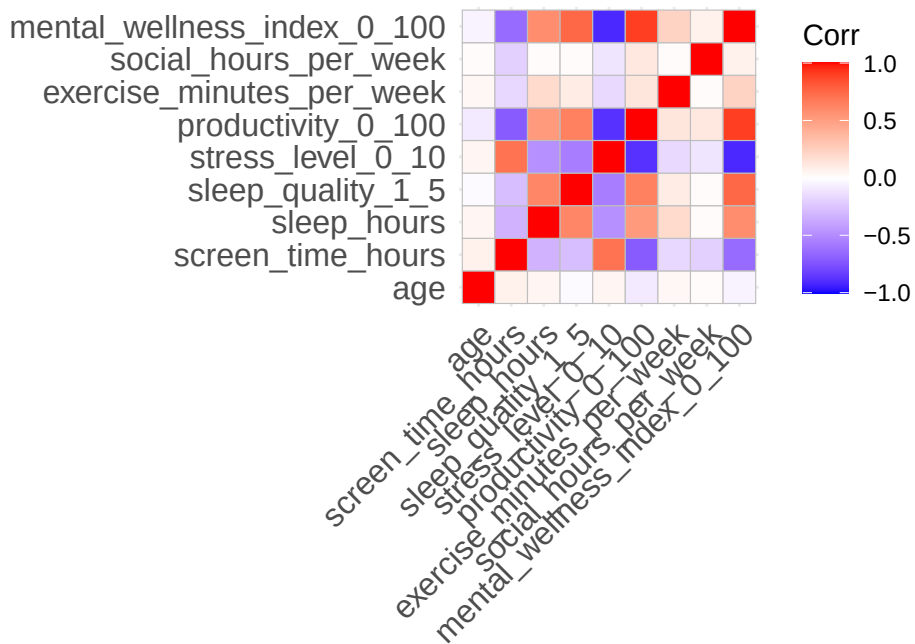
Breaking this down further, we split screen time by type (work, leisure, and total) across age groups.



Correlation Between Screen Time and Wellbeing Variables

Having established how screen time is distributed across groups, we now look at how it relates to other numeric variables in the dataset using a correlation matrix.

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.  
i Please use tidy evaluation idioms with `aes()`.  
i See also `vignette("ggplot2-in-packages")` for more information.  
i The deprecated feature was likely used in the ggcorrplot package.  
Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
```



Methodology

We plotted total screen time, leisure screen time, and work screen time against all other variables and from the plots were able to discern that across all instances that productivity, stress and mental wellness were negatively impacted due to increased screen time exposure. There appears to be some correlation between sleep hours/sleep quality and screen time as well, but this will need to be investigated further. We also created multiple models to see how significant an hour of screen time can impact mental wellness and found that one unit increase in screen time hours (one hour) decreases mental wellness by -5.2020. Additionally, we found that

social hours and the interaction term with screen time hours is significant and now one unit increase in screen time hours decreases mental wellness by -3.87844 units.

Inference & Modelling

Identifying Predictors of Screen Time

To understand what drives overall screen time, we use stepwise regression to identify the most significant predictors from the full set of available variables.

Stepwise Summary						
Step	Variable	AIC	SBC	SBIC	R2	Adj. R2
0	Base Model	1868.316	1876.299	731.119	0.00000	0.00000
1	productivity_0_100 (+)	1593.700	1605.675	457.387	0.49920	0.49790
2	sleep_quality_1_5 (+)	1550.284	1566.250	414.067	0.55295	0.55070
3	stress_level_0_10 (+)	1530.098	1550.056	393.952	0.57707	0.57380
4	work_mode (+)	1513.545	1541.486	375.639	0.59825	0.59380
5	social_hours_per_week (+)	1507.090	1539.022	369.278	0.60665	0.60080
6	occupation (+)	1483.579	1531.476	340.747	0.63645	0.62790
7	exercise_minutes_per_week (+)	1481.553	1533.442	338.931	0.64009	0.62980
8	sleep_hours (+)	1480.633	1536.514	338.213	0.64271	0.63100

Final Model Output

Model Summary			
R	0.802	RMSE	1.487
R-Squared	0.643	MSE	2.212
Adj. R-Squared	0.632	Coef. Var	16.753
Pred R-Squared	0.620	AIC	1480.633
MAE	1.176	SBC	1536.514

RMSE: Root Mean Square Error
MSE: Mean Square Error
MAE: Mean Absolute Error
AIC: Akaike Information Criteria
SBC: Schwarz Bayesian Criteria

ANOVA							
	Sum of Squares	DF	Mean Square	F	Sig.		
Regression	1591.312	12	132.609	58.013	0.0000		
Residual	884.631	387	2.286				
Total	2475.943	399					
Parameter Estimates							
	model	Beta	Std. Error	Std. Beta	t	Sig	lower
	(Intercept)	11.186	1.411		7.929	0.000	8.412
0.110	productivity_0_100	-0.087	0.012	-0.522	-7.352	0.000	-
	-0.063						
	sleep_quality_1_5	1.060	0.171	0.277	6.198	0.000	0.724
	stress_level_0_10	0.321	0.078	0.270	4.135	0.000	0.168
0.910	work_modeHybrid	-0.545	0.186	-0.106	-2.937	0.004	-
	-0.180						
	work_modeIn-person	-0.899	0.202	-0.159	-4.444	0.000	-
1.297	-0.501						
	social_hours_per_week	-0.043	0.016	-0.084	-2.730	0.007	-
	-0.012						
0.073	occupationStudent	1.465	0.343	0.261	4.275	0.000	0.791
	occupationSelf-employed	0.608	0.377	0.077	1.614	0.107	-
	1.348						
0.133	occupationEmployed	0.672	0.322	0.135	2.085	0.038	0.038
	occupationRetired	-0.083	0.504	-0.006	-0.165	0.869	-
	0.908						
1.075	exercise_minutes_per_week	-0.002	0.001	-0.054	-1.728	0.085	-
	0.000						
	sleep_hours	-0.197	0.117	-0.067	-1.684	0.093	-
0.426	0.033						

Using the variables selected by the stepwise procedure, we fit an initial model to quantify each predictor's contribution.

```
Call:
lm(formula = reformulate(predictors, response = "screen_time_hours"),
    data = st_data_without_st_overlap)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9407	-1.0134	-0.0375	1.0491	6.1067

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	11.185850	1.410794	7.929	2.39e-14	***
productivity_0_100	-0.086531	0.011770	-7.352	1.18e-12	***
sleep_quality_1_5	1.059653	0.170958	6.198	1.46e-09	***
stress_level_0_10	0.320596	0.077540	4.135	4.36e-05	***
work_modeHybrid	-0.545305	0.185642	-2.937	0.00351	**
work_modeIn-person	-0.899179	0.202336	-4.444	1.15e-05	***
social_hours_per_week	-0.042682	0.015633	-2.730	0.00662	**
occupationStudent	1.465294	0.342735	4.275	2.41e-05	***
occupationSelf-employed	0.607787	0.376574	1.614	0.10734	
occupationEmployed	0.671893	0.322299	2.085	0.03775	*
occupationRetired	-0.083266	0.504240	-0.165	0.86893	
exercise_minutes_per_week	-0.001932	0.001118	-1.728	0.08482	.
sleep_hours	-0.196534	0.116724	-1.684	0.09304	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.512 on 387 degrees of freedom

Multiple R-squared: 0.6427, Adjusted R-squared: 0.6316

F-statistic: 58.01 on 12 and 387 DF, p-value: < 2.2e-16

Impact of Screen Time on Mental Wellness

With screen time predictors established, we now model the outcome of greatest interest: mental wellness. The baseline model includes screen time, exercise, and sleep quality as predictors.

Call:

```
lm(formula = mental_wellness_index_0_100 ~ exercise_minutes_per_week +
    screen_time_hours + sleep_quality_1_5, data = cleaned_screen_time_data)
```

Residuals:

Min	1Q	Median	3Q	Max
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-27.108 -6.280 -0.895 5.382 41.585

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	23.556233	2.657715	8.863	< 2e-16 ***
exercise_minutes_per_week	0.027016	0.007149	3.779	0.000182 ***
screen_time_hours	-3.668084	0.208216	-17.617	< 2e-16 ***
sleep_quality_1_5	19.254331	0.786881	24.469	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.836 on 396 degrees of freedom

Multiple R-squared: 0.7688, Adjusted R-squared: 0.767

F-statistic: 438.8 on 3 and 396 DF, p-value: < 2.2e-16

Results

Screen time negatively impacts mental wellness. Each additional hour of screen time is associated with a 3.67 point decrease in mental wellness (on a 0–100 scale), even after controlling for exercise and sleep quality. While there is certainly a baseline amount of screen time that is “mandatory” to operate in society (i.e. texting your family, using email, navigation, etc.), the more a person can decrease their screen time, the better it will be for their mental wellbeing, generally.

Students and remote workers are most at risk for excessive screen time. The occupation and work mode plots show these groups accumulate the highest overall screen time, making them the most exposed to its negative effects. Any potential solutions to the overarching screen time problem that plagues the country might be best suited to students to derive the most impact.

Social activity can partially buffer the harmful affects of screen time. When social hours and its interaction with screen time are added to the wellness model, the negative effect of screen time shifts to -3.87 , which suggesting socialising helps but does not fully offset the impact.

Stress and productivity are also affected by screen time. The correlation matrix shows screen time has a moderate positive correlation with stress and a moderate negative correlation with productivity, supporting the broader argument beyond just mental wellness.